Abstract Class

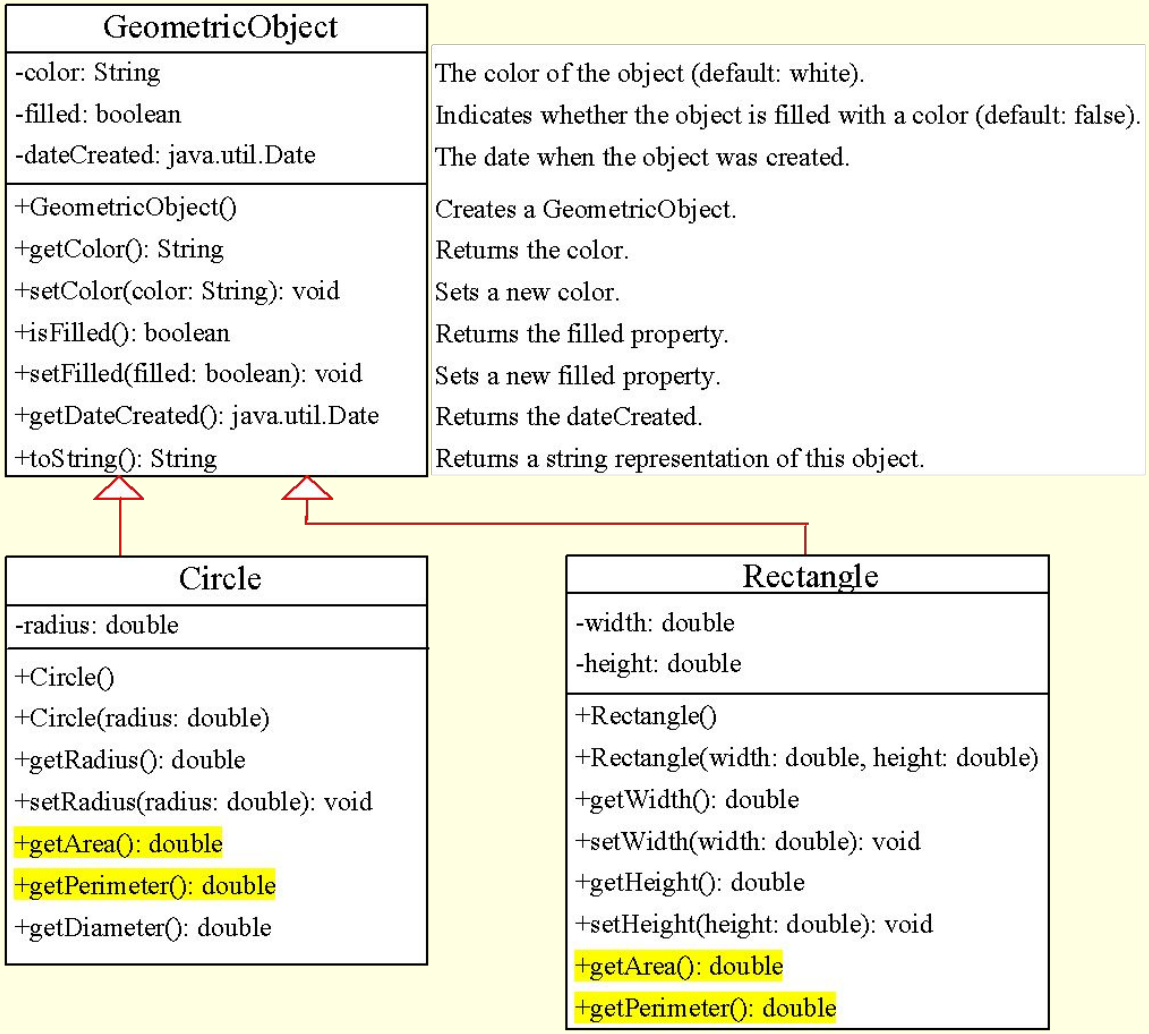
Lecture 21

Based on Slides of Dr. Norazah Yusof

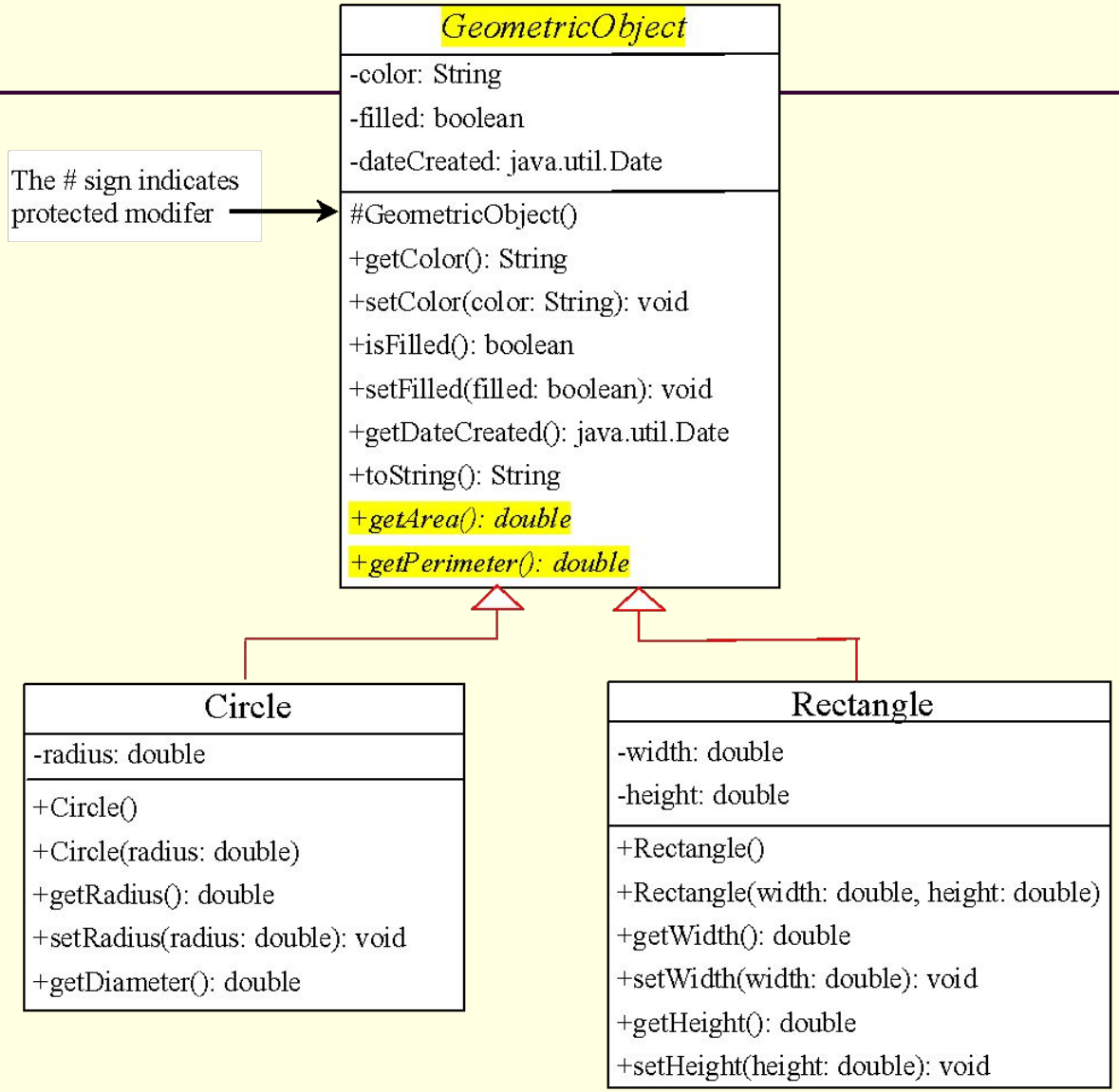
Abstract Class

- **Abstract class is a class with one or more abstract methods.**
- **The `abstract` method**
  - **Method signature without implementation**
- **An abstract class must have a modifier `abstract` included in the class heading.**
- **The `abstract` class**
  - **Cannot be instantiated**
  - **Should be extended and implemented in subclasses**

From Chapter 9



Abstract Classes



Abstract Class

```
public abstract class GeometricObject{
  private String color ="white";
  private boolean filled;
  private java.util.Date dateCreated;

  protected GeometricObject(){
    dateCreated = new java.util.Date();
  }

:
  public abstract double getArea();
  public abstract double getPerimeter();
}
```

Abstract Class

```
public class Circle extends GeometricObject
{
    private double radius;
   :
   :
    public double getArea() {
       return radius * radius * Math.PI;
    }

    public double getPerimeter() {
    return 2 * radius * Math.PI;
    }
}
```

Abstract Class

```
public class Rectangle extends GeometricObject
{
    private double radius;
   :
   :
    public double getArea() {
   return width * height;
 }

    public double getPerimeter() {
    return 2*(width + height);
    }
}
```

NOTE

- **An abstract method cannot be contained in a non-abstract class.**
- **If a subclass of an abstract superclass does not implement all the abstract methods, the subclass must be declared abstract.**
- **In other words, if a non-abstract subclass extended from an abstract class, all the abstract methods must be implemented, even if they are not used in the subclass.**

NOTE

- **An abstract class cannot be instantiated using the new operator, but you can still define its constructors, which are invoked in the constructors of its subclasses.**
- **For instance, the constructors of GeometricObject are invoked in the Circle class and the Rectangle class.**

NOTE

- **A class that contains abstract methods must be abstract.**
- **However, it is possible to declare an abstract class that contains no abstract methods.**
- **In this case, you cannot create instances of the class using the <u>new</u> operator. This class is used as a base class for defining a new subclass.**

NOTE

- **A subclass can be abstract even if its superclass is concrete.**
- **For example, the <u>Object</u> class is concrete, but its subclasses, such as <u>GeometricObject</u>, may be abstract.**

NOTE

- **A subclass can override a method from its superclass to declare it <u>abstract</u>.**
- **This is rare (unusual), but useful when the implementation of the method in the superclass becomes invalid in the subclass.**
- **In this case, the subclass must be declared abstract.**

NOTE

- **You cannot create an instance from an abstract class using the <u>new</u> operator, but an abstract class can be used as a data type.**
- **Therefore, the following statement, which creates an array whose elements are of <u>GeometricObject</u> type, is correct.**

```
GeometricObject[] geo = new GeometricObject[10];
```

Example: Using the GeometricObject Class

- **Objective: This example creates two geometric objects: a circle, and a rectangle, invokes the equalArea method to check if the two objects have equal area, and invokes the displayGeometricObject method to display the objects.**

Exercises

- Self tests
  - Question 1, page 174
  - Question 2, page 175
  - Question 3, page 176
  - Question 1, page 192
- Lab Exercises:
  - Question 2, page 177
  - Question 2, page 192